DEPARTMENT OF AGRICULTURE IV-A · DEPDev IV-A

Food Insecurity Risk Forecast

Rizal · Q4 2025 (Oct-Dec)

Generated 29 September 2026 from aiPHeed model output.

What this number is

The modelled probability that this area falls in the HIGH class of the composite food stress label. **It is an indicative relative risk estimate, not an official food insecurity classification**, and it is published at the Maximum Reliable Lead Time the model established empirically.

Headline

FOOD INSECURITY RISK PROBABILITY

0.528

HIGH · quarter-on-quarter +0.0000

Risk class	**HIGH**
Rank in CALABARZON	**2 of 5**
Population (2020 CPH)	**3,330,143**
Poverty incidence (PSA 2021)	**3.7%**
Articles analysed	**0**

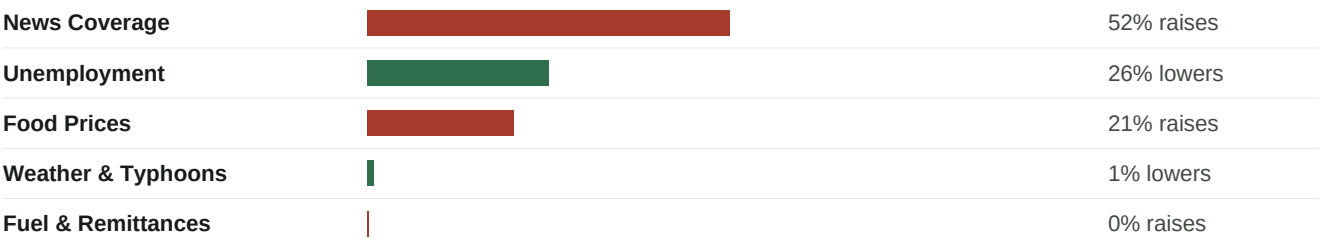
Limited signal

Only 0 relevant news articles were geocoded to this province for the quarter. News-derived drivers should be treated as weak evidence.

Forecast horizon

Same-quarter estimate. No forward lead time met the 75% reliability threshold of Section 3.4.4, so none is published.. Indicative relative risk only. Walk-forward accuracy did not reach 75% at any forward horizon, so this output is reported as a same-quarter estimate and not as a forecast.

What is driving the figure



Red bars raise the predicted shortfall, **green** lower it. A large bar is not automatically bad news -- share of the explanation and direction of effect are separate things.

For Rizal in 2025-Q4, the model's prediction breaks down as: News Coverage 52%, Unemployment 26%, Food Prices 21%, Weather & Typhoons 1%, Fuel & Remittances 0%. News Coverage is the largest single contributor, pushing the score up. News Coverage and Food Prices clear the 20% line while raising the score. Unemployment clears the 20% line while lowering it. Shares are SHAP contributions to the province's food insecurity risk probability, recomputed every quarter.

Municipal distribution

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). **They are not independent municipal forecasts** -- every municipality in a province rises and falls with it.

#	City / Municipality	Class	Population	Poverty	Index
1	**Rodriguez**	Municipality	443,954	6.5%	0.3595
2	**Taytay**	Municipality	386,451	3.5%	0.2400
3	**Cainta**	Municipality	376,933	3.5%	0.2128
4	**Angono**	Municipality	130,494	3.5%	0.1264
5	**San Mateo**	Municipality	273,306	3.5%	0.1260
6	**Binangonan**	Municipality	313,631	3.5%	0.1207
7	**Teresa**	Municipality	64,072	3.5%	0.0914
8	**Cardona**	Municipality	50,143	3.5%	0.0722
9	**Morong**	Municipality	71,151	3.5%	0.0561
10	**Antipolo**	City	887,399	3.2%	0.0502
11	**Baras**	Municipality	87,637	3.5%	0.0364
12	**Pililla**	Municipality	71,535	3.5%	0.0362
13	**Jala-Jala**	Municipality	34,017	3.5%	0.0305
14	**Tanay**	Municipality	139,420	3.5%	0.0288

News evidence

No relevant articles were geocoded to this subject for the quarter.

Method

The model predicts, for each CALABARZON province, the probability that the province falls in the HIGH class of the composite food stress label -- SWS quarterly hunger prevalence combined with the province's food CPI deviation from the regional mean, thresholded at the global median. Inputs are the bias-corrected Food Stress Sentiment Index and its lags and acceleration, the five bilingual trigger proportions, and government indicators from PSA, BSP, DOE, PAGASA and PhilRice.

Measure	Value
Maximum Reliable Lead Time	0 quarter(s)
Accuracy	0.6000
Precision (weighted)	0.9200
Recall (weighted)	0.6000
F1 score (weighted)	0.6762
AUC-ROC	0.9444
F1 target (>= 0.75)	FAIL
AUC-ROC target (>= 0.80)	PASS
Features in the model	12
Baseline: aipheed lgbm combined	0.6727
Baseline: full feature logistic	0.5455
Baseline: lgbm nlp only	0.6182
Baseline: lgbm psa only	0.7273
Baseline: naive persistence	0.8000
Baseline: nlp only logistic	0.5455
Baseline: psa only logistic	0.5818

Accuracy must be read against its baselines. The comparison models above are evaluated on the same walk-forward folds at the same forecast gap, so the figures are directly comparable. Naive persistence in particular is a demanding baseline on an autocorrelated label: a model that does not clearly beat it is not earning its complexity.

Limitations

Indicative risk, not official classification

Outputs are relative ordinal estimates of potential risk. They are not definitive determinations of food insecurity status and must not be read as official government assessments or policy recommendations.

Proxy label, not a direct measurement

No publicly accessible province-quarter food insecurity label exists in the Philippines. The training target is a composite of SWS hunger prevalence, which has no province-level resolution, and PSA food CPI deviation, which reflects supply-side price conditions rather than household hunger experience directly.

Limited news signal

Only 0 relevant articles were geocoded to this subject for the quarter, below the threshold for a dependable news component. The news-derived drivers carry correspondingly little weight.

Municipal figures are disaggregated, not modelled

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). Poverty values are largely province means rather than municipal Small Area Estimates, so within-province ordering is indicative only.

Reviewed before publication

Province-quarter figures pass through analyst review. A withheld figure is excluded from this report and from the regional average.

Scope of this assessment

This assessment reports findings only. It does not recommend interventions: no validated mapping exists between this indicator and a response protocol. Pair it with the Department's own standing guidance.